Incident Handler's Journal

[Your Name] | Google Cybersecurity Professional Certificate — Course 6: Sound the Alarm: Detection and Response

This journal documents incident investigations and hands-on practice with cybersecurity tools completed throughout the course. Each entry records what happened, which tools were used, and the 5 W's of an incident (who, what, when, where, why), mapped to the NIST Incident Response Lifecycle where applicable.

Date: [Date]	**Entry:** #1 — Ransomware attack on a health care clinic
Description	Documented a ransomware incident at a small U.S. primary-care clinic that disrupted business operations. This entry covers the **Detection and Analysis** phase of the NIST Incident Response Lifecycle: the clinic detected the encryption of its files and began analyzing how the attackers gained access.
Tool(s) used	None.
The 5 W's	• **Who:** An organized group of unethical hackers known to target health care and transportation organizations. • **What:** A ransomware attack encrypted critical files; a ransom note demanded payment in exchange for the decryption key. • **When:** Tuesday, at approximately 9:00 a.m. • **Where:** At the small U.S. health care clinic, affecting employees' computers and patient-care systems. • **Why:** Attackers sent targeted phishing emails with a malicious attachment. Once an employee opened it, malware was installed, giving the attackers access to deploy ransomware.
Additional notes	How could the clinic prevent this in the future (security awareness training, email filtering, offline backups)? Should the clinic pay the ransom? Paying does not guarantee recovery and may fund further attacks.

Date: [Date]	**Entry:** #2 — Analyzing network traffic with Wireshark
Description	Used Wireshark for the first time to open and analyze a packet capture file. Practiced applying display filters, inspecting individual packets, and reading protocol details across the network layers.
Tool(s) used	• **Wireshark** — an open-source network protocol analyzer with a graphical user interface. • It captures and displays network traffic so analysts can investigate suspicious activity packet by packet. • Filters (e.g., by IP address, protocol, or port) help isolate relevant traffic quickly. • Valuable for detecting malicious traffic and understanding how an intrusion occurred.
The 5 W's	N/A — this entry documents tool usage rather than a specific incident investigation.
Additional notes	The interface was overwhelming at first, but filters made the data much easier to navigate. I want to keep practicing so I can recognize normal versus abnormal traffic patterns faster.

Date: [Date]	**Entry:** #3 — Capturing network traffic with tcpdump
Description	Used tcpdump in a Linux terminal to identify network interfaces, capture live traffic, save it to a .pcap file, and then filter and read the captured packet data from the command line.
Tool(s) used	• **tcpdump** — a command-line network protocol analyzer. • Lightweight and available on most Linux systems, making it useful on servers without a GUI. • Captures traffic to a file (-w) that can later be read (-r) with tcpdump or Wireshark. • Supports filters by host, port, and protocol to focus on relevant packets.
The 5 W's	N/A — this entry documents tool usage rather than a specific incident investigation.
Additional notes	Working entirely in the command line was challenging, but it reinforced how network traffic is structured. Knowing both tcpdump and Wireshark gives flexibility in different environments.

Date: [Date]	**Entry:** #4 — Investigating a suspicious file hash
Description	Analyzed a SHA256 file hash from a suspicious email attachment using VirusTotal and determined the file was malicious. This entry covers the **Detection and Analysis** phase: the IDS detected the file and I analyzed its indicators of compromise (IoCs).
Tool(s) used	• **VirusTotal** — an online service that analyzes files, URLs, domains, and IP addresses against many security vendors' detections. • Used to confirm the hash was flagged as malicious (identified as Flagpro malware, associated with the BlackTech group). • Community insights provided additional IoCs such as related domains, IP addresses, and file behavior. • Applied the **Pyramid of Pain** to classify the IoCs found.
The 5 W's	• **Who:** An unknown malicious actor who sent the phishing email. • **What:** An employee received an email with a password-protected spreadsheet attachment. After opening it with the password in the email, a malicious payload executed on their computer. • **When:** The email arrived at 1:11 p.m.; the file was opened at 1:13 p.m.; executables ran at 1:15 p.m.; the IDS alert fired at 1:20 p.m. • **Where:** On an employee's computer at a financial services company. • **Why:** The employee was deceived by a phishing email and opened the malicious attachment.
Additional notes	How can the organization improve security awareness so employees are more careful with unexpected attachments? Should password-protected attachments from external senders be blocked by default?

Date: [Date]	**Entry:** #5 — Responding to a phishing alert with a playbook
Description	Followed the organization's phishing playbook to respond to the alert from Entry #4. Evaluated the email's indicators (sender, attachment, message content), updated the alert ticket, and escalated it to a Level 2 SOC analyst. This covers the **Detection and Analysis** phase and prepares for **Containment**.
Tool(s) used	• **Phishing playbook and flowchart** — a documented process for handling phishing alerts consistently. • **Alert ticketing system** — used to record findings, update the status, and escalate the ticket.
The 5 W's	• **Who:** A malicious sender using a suspicious email address and a misspelled, inconsistent sender name. • **What:** A phishing email delivered a malicious attachment (the file hash confirmed as malicious in Entry #4). • **When:** The same day as the alert, starting at 1:11 p.m. • **Where:** The employee's corporate email inbox and workstation. • **Why:** The attacker used social engineering (posing as a job applicant) to trick the employee into opening the file.
Additional notes	Playbooks make response faster and more consistent. I noticed several red flags in the email (grammar errors, sender mismatch) that training could help employees spot.

Date: [Date]	**Entry:** #6 — Reviewing a final report: data theft incident
Description	Reviewed a final incident report for a data breach at a retail company. This covers the **Post-Incident Activity** phase, where the organization documented the root cause, impact, and lessons learned.
Tool(s) used	None.
The 5 W's	• **Who:** An unknown attacker who exploited the company's e-commerce web application. • **What:** Customer personally identifiable information (PII) and financial data were stolen, and the attacker demanded a ransom. • **When:** The first extortion email was received on December 22; a second email with stolen sample data followed on December 28. • **Where:** The company's e-commerce web application. • **Why:** A forced browsing vulnerability: the attacker modified the order number in the URL of the purchase confirmation page to access other customers' data.
Additional notes	Recommendations included routine vulnerability scans, penetration testing, allowlisting of URLs, and restricting access to authenticated users only. Post-incident reviews turn a costly breach into lessons that prevent the next one.

Date: [Date]	**Entry:** #7 — Exploring signatures and logs with Suricata
Description	Used Suricata to examine a custom IDS rule, trigger it against a packet capture, and analyze the resulting alerts in fast.log and eve.json.
Tool(s) used	• **Suricata** — an open-source intrusion detection system (IDS), intrusion prevention system, and network analysis tool. • Rules (signatures) are made of an action, a header, and rule options. • Generates alerts in fast.log and detailed JSON events in eve.json. • Used jq to parse eve.json output efficiently.
The 5 W's	N/A — this entry documents tool usage rather than a specific incident investigation.
Additional notes	Writing and reading rules showed me how detection logic works behind the alerts analysts receive. Well-tuned signatures reduce false positives.

Date: [Date]	**Entry:** #8 — Performing queries with Splunk
Description	Uploaded sample log data to Splunk Cloud and used Search Processing Language (SPL) queries to investigate failed SSH login attempts for the root account on a mail server.
Tool(s) used	• **Splunk Cloud** — a SIEM tool that collects, indexes, and analyzes log data. • SPL queries filter events by index, host, and keywords (e.g., index=main host=mailsv fail* root). • Helps analysts quickly identify patterns such as brute force login attempts.
The 5 W's	N/A — this entry documents tool usage rather than a specific incident investigation.
Additional notes	Splunk made searching large amounts of log data much faster than reviewing logs manually. I want to learn more advanced SPL commands and dashboards.

Date: [Date]	**Entry:** #9 — Performing queries with Chronicle
Description	Used Google Security Operations (Chronicle) to investigate a suspicious phishing domain, identify which assets accessed it, and review the resolved IP addresses and POST requests to determine whether credentials may have been submitted.
Tool(s) used	• **Chronicle (Google SecOps)** — a cloud-native SIEM tool. • Domain searches show affected assets, resolved IP addresses, and event timelines. • Threat intelligence data (e.g., VirusTotal context) helps confirm whether a domain is malicious. • Used to identify POST requests to a login page, suggesting possible credential theft.
The 5 W's	N/A — this entry documents tool usage rather than a specific incident investigation.
Additional notes	Chronicle's visual timeline made it easy to pivot between domains, IPs, and assets. Comparing it with Splunk helped me understand different SIEM workflows.

Reflections / Notes

Were there any specific activities that were challenging for you? Why or why not?

The tcpdump activity was the most challenging because I had to work entirely in the Linux command line and remember the syntax for options and filters. Reading raw packet output was difficult at first, but repeated practice helped me understand how network traffic is structured and how to interpret it.

Has your understanding of incident detection and response changed since taking this course?

Yes. Before this course, I thought incident response was mainly about stopping attacks. Now I understand it is a structured lifecycle that includes preparation, detection and analysis, containment, eradication, recovery, and post-incident review. Documentation and playbooks are essential at every phase.

Was there a specific tool or concept that you enjoyed the most? Why?

I enjoyed working with SIEM tools like Splunk and Chronicle the most. They made it possible to search through large amounts of log data quickly and connect events into a clear picture of an incident. I want to continue building my skills with these tools.